Example 4.22 $\frac{d\underline{x}}{dt} + \underline{\underline{P}}(t)\underline{x} = \underline{g}(t)$

$$\underline{\underline{P}} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \qquad \underline{g} = \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix}$$

$$= \underline{\underline{A}}$$

Homogeneous problem

$$|\underline{\underline{A}} - r\underline{\underline{I}}| = 0 \qquad \begin{vmatrix} -2-r & 1 \\ 1 & -2-r \end{vmatrix} = 0 \Rightarrow (-)^2(2+r)^2 - 1 = 0$$

$$3 + r^2 + 4r = 0$$

$$(r+3)(r+1) = 0$$

the eigenvalues are $r_1 = -3$, $r_2 = -1$

Eigenvectors

- $r_1 = -3$ $\quad (\underline{\underline{A}} - r_1\underline{\underline{I}})\underline{v}^{(1)} = \underline{0} \Rightarrow \begin{pmatrix} -2-(-3) & 1 \\ 1 & -2-(-3) \end{pmatrix}\begin{pmatrix} v_1^{(1)} \\ v_2^{(1)} \end{pmatrix} = \underline{0}$

$$\Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \underline{0}$$

$$\therefore \underline{v}^{(1)} = \begin{pmatrix} \beta \\ -\beta \end{pmatrix}$$ where β is any non-zero constant.

- $r_2 = -1$ $\quad \begin{pmatrix} -2-(-1) & 1 \\ 1 & -2-(-1) \end{pmatrix}\begin{pmatrix} v_1^{(2)} \\ v_2^{(2)} \end{pmatrix} = \underline{0}$

$$\Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} v_1^{(2)} \\ v_2^{(2)} \end{pmatrix} = \underline{0}$$

$$\Rightarrow \underline{v}^{(2)} = \begin{pmatrix} \alpha \\ \alpha \end{pmatrix}$$ where α is any non-zero constant.

$$W[\underline{x}^{(1)}, \underline{x}^{(2)}] = \begin{vmatrix} \beta e^{-3t} & \alpha e^{-t} \\ -\beta e^{-3t} & \alpha e^{-t} \end{vmatrix} = 2\alpha\beta e^{-4t} \neq 0$$

because $e^{-4t} \neq 0$ and $\alpha\beta \neq 0$ because $\alpha \neq 0$ and $\beta \neq 0$.

$\therefore \underline{x}^{(1)}$ and $\underline{x}^{(2)}$ form a fundamental set.

Solution of the homogeneous problem is

$$\underline{x}_c = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} \qquad -\infty < t < \infty$$

Nonhomogeneous problem $\underline{g} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} t$

we seek a particular solution of the form

$\underline{x}_p = \underline{a}\, t e^{-t} + \underline{b}\, e^{-t} + \underline{c}\, t + \underline{d}$ "trial solution"

extra — polynomial

because $r_2 = -1$ a single root of the characteristic polynomial

Substitute $\underline{x}_p$ into linear system:

$$\text{LHS}: \frac{d\underline{x}_p}{dt} = \underline{a}\, t e^{-t}(-1) + e^{-t}\underline{a} + \underline{b} e^{-t}(-1) + \underline{c}$$

$$\text{RHS}: \underline{A}\underline{x}_p + \underline{g} = \underline{A}\left[\underline{a}\, t e^{-t} + \underline{b} e^{-t} + \underline{c}\, t + \underline{d}\right] + \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} t$$

Equate LHS = RHS to find $\underline{a}, \underline{b}, \underline{c}$ & $\underline{d}$

(1) Coeff $t e^{-t}$ $\quad -\underline{a} = \underline{A}\,\underline{a}$

(2) Coeff e^{-t} $\quad \underline{a} - \underline{b} = \underline{A}\,\underline{b} + \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

(3) Coeff t $\quad \underline{0} = \underline{A}\,\underline{c} + \begin{pmatrix} 0 \\ 3 \end{pmatrix}$

} 4 unknowns $\underline{a}, \underline{b}, \underline{c}$ & $\underline{d}$ and 4 equations.

(4) constants $\quad \underline{c} = \underline{A}\,\underline{d}$

Solving

(1) $(\underline{A} + \underline{I})\underline{a} = \underline{0} \Rightarrow \begin{pmatrix} -2+1 & 1 \\ 1 & -2+1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \underline{0}$

is satisfied by $\underline{a} = \begin{pmatrix} \gamma \\ \gamma \end{pmatrix}$

(2) requires $\begin{pmatrix} \gamma \\ \gamma \end{pmatrix} - \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

solve $\underline{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_1 - 1 \end{pmatrix} = b_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for any b_1

Take $b_1 = 0$ to obtain $\underline{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$ //

(3) $\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \end{pmatrix} \Rightarrow \underline{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

(4) $\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \Rightarrow \underline{d} = \begin{pmatrix} -4/3 \\ -5/3 \end{pmatrix}$

Hence we have the particular solution

$$\underline{x}_p = \begin{pmatrix} 1 \\ 1 \end{pmatrix} t e^{-t} + \cancel{\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{-t} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} t + \begin{pmatrix} -4/3 \\ -5/3 \end{pmatrix}$$

$\underline{a}$ ↑ confirm — $\underline{b}$

General solution

$$\underline{x}(t) = c_1\begin{pmatrix}1\\-1\end{pmatrix}e^{-3t} + c_2\begin{pmatrix}1\\1\end{pmatrix}e^{-t} + \begin{pmatrix}1\\1\end{pmatrix}te^{-t} + \begin{pmatrix}0\\-1\end{pmatrix}e^{-t} + \begin{pmatrix}1\\2\end{pmatrix}t + \begin{pmatrix}-4/3\\-5/3\end{pmatrix} \qquad -\infty < t < \infty$$

Coursework 1 – now to midnight (40% mins)